Risk of Stroke/Transient Ischemic Attack or Myocardial Infarction with Herpes Zoster: A Systematic Review and Meta-Analysis

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Background: Accumulating evidence indicates that herpes zoster (HZ) may increase the risk of stroke/transient ischemic attack (TIA) or myocardial infarction (MI), but the results are inconsistent. We aim to explore the relationship between HZ and risk of stroke/TIA or MI and between herpes zoster ophthalmicus (HZO) and stroke. *Methods:* We estimated the relative risk (RR) and 95% confidence intervals (CIs) with the meta-analysis. Cochran's Q test and Higgins I^2 statistic were used to check for heterogeneity. *Results:* HZ infection was significantly associated with increased risk of stroke/TIA (RR = 1.30, 95% CI: 1.17-1.46) or MI (RR = 1.18, 95% CI: 1.07-1.30). The risk of stroke after HZO was 1.91 (95% CI 1.32-2.76), higher than that after HZ. Subgroup analyses revealed increased risk of ischemic stroke after HZ infection but not hemorrhagic stroke. The risk of stroke was increased more at 1 month after HZ infection than at 1-3 months, with a gradual reduced risk with time. The risk of stroke after HZ infection was greater with age less than 40 years than 40-59 years and more than 60 years. Risk of stroke with HZ infection was greater without treatment than with treatment and was greater in Asia than Europe and America but did not differ by sex. *Conclusions:* Our study indicated that HZ infection was associated with increased risk of stroke/TIA or MI, and HZO infection was the most marked risk factor for stroke. Further studies are needed to explore whether zoster vaccination could reduce the risk of stoke/TIA or MI. **Key Words:** Stroke—myocardial infarction—herpes zoster—risk—meta-analysis.
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Introduction

Stroke is classically characterized as a neurological deficit attributed to an acute focal injury of the central nervous system by a vascular cause.[1] All over the world, stroke, the second most common cause of death, has become a significant public health issue. Stroke occurs in about 15 million people worldwide per year, leading to 5 million deaths and an additional 5 million people disabled.[2,3] Atherosclerosis and embolism are the most important causes of stroke. Common risk factors include smoking, atrial fibrillation, diabetes, and hypertension. Infection-related stroke is differentiated from stroke due to other reasons because some infectious causes can be treated with antibiotics and prevented by vaccination for further control of stroke.

Myocardial infarction (MI) is also a major cause of morbidity and mortality worldwide.[3] Increased MI events after acute infection are thought to be related to endothelial dysfunction, characterized by atheromatous plaque rupture and development of a prothrombotic environment.[4]

Herpes zoster (HZ) infection is a significant public health problem in aging populations, annually affecting 1 million Americans and more than 88,650 immunocompetent people older than 60 years in the United Kingdom.[5-7] HZ, also known as shingles, is a painful vesicular eruption that is typically restricted to 1 or 2 dermatomes and is induced by the reactivation of dormant varicella zoster virus (VZV), a ubiquitous pathogen that persists asymptomatically in the sensory ganglia after primary chickenpox in childhood.[8] HZ infection occurs in people with diminished cell-mediated immunity. HZ infection is often pernicious, with frequent complications, the most common being postherpetic neuralgia leading to long-term pain, depression, and disability.[9] Other severe complications include vasculopathy, giant cell arteritis, myelitis, encephalitis, meningoencephalitis, keratitis, ocular involvement, cranial nerve palsies, and motor nerve involvement.[10-13]

Recent studies have proposed that the risk of stroke or MI may be increased in the year following an episode of HZ, possibly also mediated by VZV replication in cerebral arterial walls. The infection is spread along the nerve fibers to the blood vessels and thrombotic responses are induced, thereby resulting in cerebral vasculopathy.[14,15] In our analysis, we classified transient ischemic attack (TIA) as minor stroke because of the similar pathogenic mechanism. However, the relationship between HZ infection and risk of stroke/TIA or MI is still unclear. A positive association was revealed by some epidemiologic studies,[16-21] but others indicated negative results.[8,22-25]

We performed a systematic review and meta-analysis to summarize the published epidemiologic evidence on the relationship between HZ infection and risk of stroke/TIA or MI or between herpes zoster ophthalmicus (HZO) infection and stroke. We also evaluated the association between HZ infection and stroke by time since HZ, age, sex, ethnicity status, and treatment status.

Materials and Methods

Search Strategy and Study Selection

Two authors (Y.Z. and G.L.) independently searched PubMed for articles published in English up to October 20, 2016, by using the key words ("Stroke"[Mesh] OR Stroke[tiab] OR ischemic stroke[tiab] OR ischaemic stroke[tiab] OR non-ischemic stroke[tiab] OR hemorrhagic stroke[tiab] OR poststroke[tiab] OR apoplexy[tiab] OR cerebral arterial disease[tiab] OR cerebrovascular accident[tiab] OR "Myocardial Infarction"[Mesh] OR myocardial infarction[tiab] OR myocardium infarction[tiab] OR myocardic infarction[tiab] OR "Ischemic Attack, Transient"[Mesh] OR transient ischaemic attack[tiab] OR transient ischemic attack[tiab] OR ischemic attack*[tiab] OR ischaemic attack*[tiab] OR brain ischemi*[tiab] OR brain ischaemi*[tiab] OR transient brain ischemia[tiab] OR cerebral ischemia[tiab]), AND ("Herpes Zoster"[Mesh] OR Herpes Zoster[tiab] OR Shingles[tiab] OR Zona[tiab] OR Zoster[tiab] OR VZV[tiab]). Titles, abstracts, full texts, and reference lists of all identified reports were reviewed in duplicate by the 2 authors, and extracted articles were double-checked by K.L. Disagreements were resolved by discussion among the 3 authors (Y.Z., G.L., and K.L.). Reference lists from related main studies and review articles were checked for additional relevant reports.

Inclusion and Exclusion Criteria

Articles were considered eligible if (1) they included peer-reviewed data from an observational study; (2) the criteria for confirmation of the HZ diagnosis in our included studies were as follows: (1) physician's diagnosis of HZ with clinical symptoms (vesicular rash or dermatomal pain or others); (2) serological analysis (dried blood spot analysis) or laboratory tests (Tzanck smear or viral culture or polymerase chain reaction); and (3) HZ-related diagnostic codes with the International Classification of Diseases, Tenth Revision (ICD-10: B02), or Ninth Revision (ICD-9: 053). Only the cases satisfied with the above 3 criteria in the meantime were included in our studies. Thus, those cases with vesicle symptoms not caused by VZV infection were excluded in a stringent manner. Articles were also considered eligible if (3) stroke was diagnosed with clinical information and radiological investigation. Stroke was identified using diagnostic codes with ICD-10 I60-64 and G45 or ICD-9 430-438; (4) concerned the risk of stroke/TIA or MI after HZ infection; and (5) reported multivariate-adjusted relative risks (RRs) and 95% confidence intervals (CIs). We excluded (1) studies of patients with a diagnosis of stroke before HZ diagnosis; (2) reviews and letters; (3) duplicate publications; (4) non-English language articles; and (5) data insufficient for calculating a risk estimate with 95% CIs, or unqualified data. When 2 articles appeared to report results with overlapping data, only the data representing the most recent

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publication or with the larger sample size were included in the meta-analysis. No publications were excluded on the basis of quality, sample size, or other objective criteria relevant to study design and analysis.

Quality Assessment

We evaluated the quality of reports by the Newcastle-Ottawa quality assessment scale[26] (http://www.ohri.ca/programs/clinical_epidemiology/nosgen.pdf). Two independent reviewers read and scored all included studies. The Newcastle-Ottawa scale contains 8 items, 7 applicable to our study question. Items were classified into 3 domains: selection, comparability, and outcome. The total scores of 0-3, 4-6, and 7-9 stars were assigned for low, moderate, and high qualities, respectively. Any discrepancies between the 2 reviewers were resolved by joint reassessment until consensus was reached.

Data Extraction

We extracted data on (1) publication details (first author's name, year of publication, and study design); (2) baseline characteristics of the study population (country, study period, number of observation groups, age, follow-up time); and (3) other details (infection types, endpoints). If these data were absent, we sent an email to the authors. We used the results of the original studies from multivariable models with the most complete adjustment for potential confounders.

Statistical Analysis

We extracted the RRs and their 95% CIs from studies to evaluate the risk of stroke/TIA or MI after HZ infection or risk of stroke after HZO infection. A pooled RR higher than 1 indicated greater risk for the group. The effect of stroke/TIA or MI after HZ exposure was considered statistically significant with 95% CIs for the pooled RR not overlapping 1 in the forest plot. Cochran's Q test and Higgins I^2 statistic were used to check for heterogeneity of the combined result.[27,28] Studies were considered heterogeneous with $P_{heterogeneity}$ less than .1 or I^2 higher than 50%. With heterogeneity, the random effects model (DerSimonian–Laird method) was used[29]; otherwise, the fixed effects model (Mantel–Haenszel method) was used.[30] We investigated potential sources of heterogeneity by subgroup analyses of time since HZ, area, age, sex, and treatment. Sensitivity analyses involved excluding studies one by one. We investigated publication bias by the Egger's linear regression test[31] and Begg's methods.[32] The effect of potential publication bias on risk assessment was further evaluated by the Duval and Tweedie nonparametric "trim-and-fill" method.[33] Two-sided $P < .05$ was considered statistically significant. All analyses involved use of STATA 13.0 (STATA Corp., College Station, TX).

Results

Literature Search and Study Characteristics

We identified 392 relevant studies in the literature search; 20 were removed as duplicates and 331 were excluded on the basis of titles and abstracts. After manually screening the full text of 41 articles, 30 articles were excluded. Finally, our analysis included 11 studies[8,16-25] (Fig 1). In our included studies, stroke included ischemic and hemorrhagic strokes. Furthermore, intracerebral hemorrhage or subarachnoid hemorrhage was included in hemorrhagic stroke. Five studies report the RR of ischemic stroke and 3 studies report the RR of hemorrhagic stroke, while the rest report the RR of stroke of both subtypes. Among the 11 eligible studies, 3 were conducted in Asia, 3 in America, 4 in Europe, and 1 was an international collaboration. The articles were published between 2009 and 2016. Three were case–control studies, and the remaining were cohort studies. Our meta-analysis included 4,620,980 participants and the mean follow-up duration ranged from 1 to 12.5 years. The basic characteristics of included studies are shown in Table 1. The overall quality of the studies was high (median score 7.8/9, range 7-9); the detailed Newcastle-Ottawa quality scores are shown in Table S1. The studies varied in terms of primary outcome: 10 articles reported on HZ infection and stroke, 1 article on TIA, 3 articles on HZ infection and MI, and 5 articles on HZO infection and stroke.

Risk of Stroke/TIA or MI after HZ or HZO

Figure 2 shows the results of HZ infection and stroke, suggesting heterogeneity ($I^2 = 81.9\%$, $P = .001$), which rationalized further exploration to disclose factors leading to the heterogeneity. The pooled RR was 1.30 (95% CI 1.17-1.46, $P_{heterogeneity} < .001$) by the random effects model, which suggested HZ associated with risk of stroke.

Figure 3 shows the results of HZ infection and MI, suggesting heterogeneity ($I^2 = 90.3\%$, $P = .001$). The pooled RR was 1.18 (95% CI 1.07-1.30, $P_{heterogeneity} < .001$) by the random effects model, which suggested HZ associated with risk of MI.

Figure 4 shows the results of HZO infection and stroke, suggesting heterogeneity ($I^2 = 84.4\%$, $P = .001$). The pooled RR was 1.91 (95% CI 1.32-2.76, $P_{heterogeneity} < .001$) by the random effects model, which suggested HZO associated with risk of stroke.

Subgroup Analyses

Subgroup analyses revealed increased risk of ischemic stroke after HZ infection (RR = 1.22, 95% CI 1.07-1.40; $P_{heterogeneity} = .097$) and hemorrhagic stroke after HZ infection with no statistical significance (RR = 1.52, 95% CI .88-2.63; $P_{heterogeneity} = .001$). Subgroup analyses revealed risk of stroke associated with time since HZ infection: at 1 month (RR = 1.92, 95% CI 1.47-2.51; $P_{heterogeneity} = .001$), 1-3 months (RR = 1.27, 95% CI 1.05-1.53; $P_{heterogeneity} = .026$), and more

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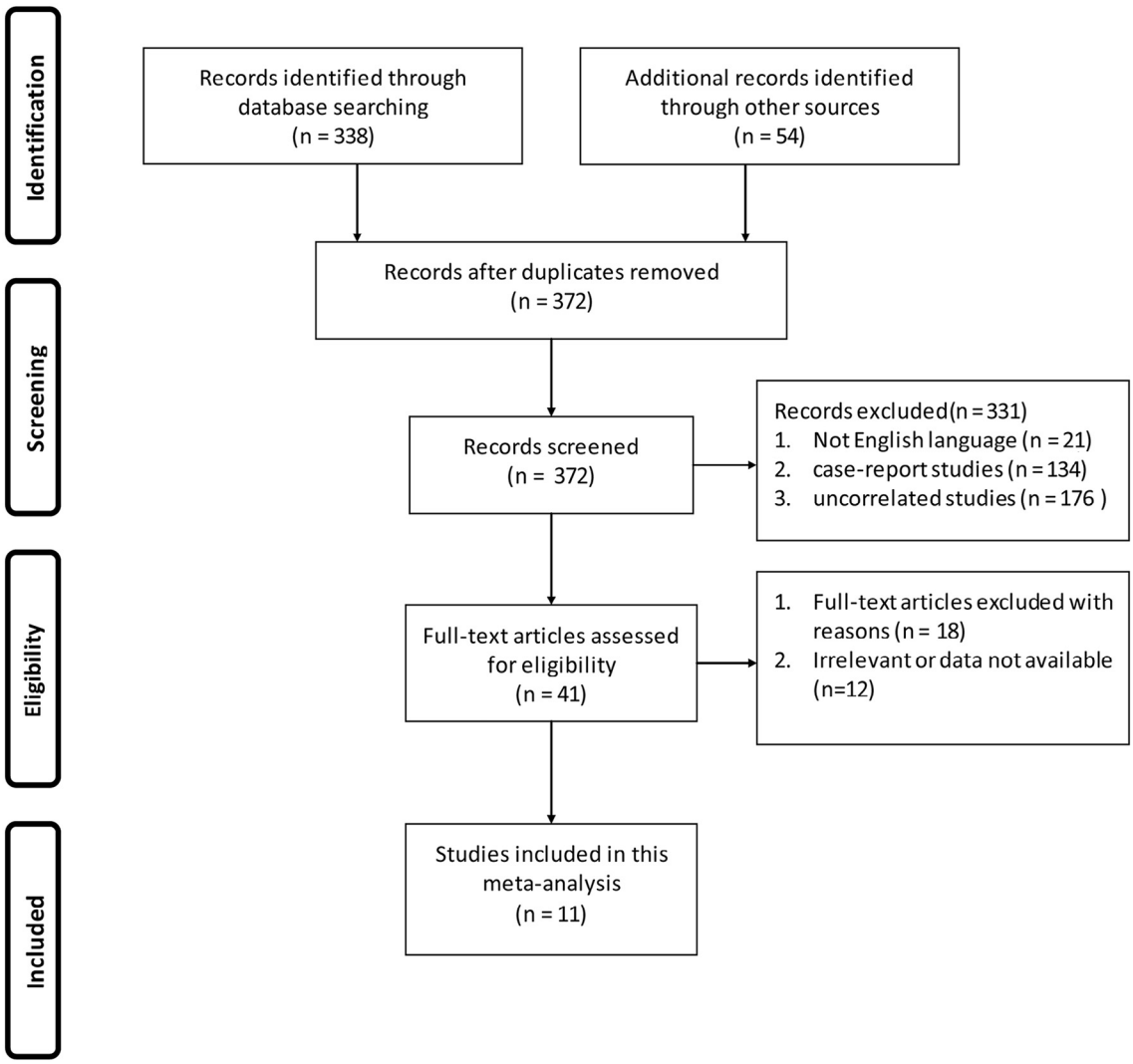

Figure 1. *Flow diagram of the search result for the meta-analysis.*

than 12 months (RR = 1.05, 95% CI 1.02-1.09; $P_{heterogeneity}$ = .623) (Table 2). Risk of stroke was increased with HZ infection by age: less than 40 years (RR = 2.03, 95% CI 1.64-2.51; $P_{heterogeneity}$ = .001), 40-59 years (RR = 1.35, 95% CI 1.20-1.53; $P_{heterogeneity}$ = .001), and more than 60 years (RR = 1.32, 95% CI 1.17-1.48; $P_{heterogeneity}$ = .001). Risk of stroke with HZ infection was associated more with no treatment (RR = 1.38, 95% CI 1.06-1.80; $P_{heterogeneity}$ = .001) than with treatment (RR = 1.17, 95% CI 1.08-1.27; $P_{heterogeneity}$ = .001) and was higher in Asia (RR = 1.61, 95% CI 1.11-2.30; $P_{heterogeneity}$ = .001) than Europe (RR = 1.23, 95% CI 1.13-1.34; $P_{heterogeneity}$ = .001) and America (RR = 1.28, 95% CI 1.07-1.53; $P_{heterogeneity}$ = .001) but did not differ by sex (men: RR = 1.31, 95% CI 1.09-1.57; $P_{heterogeneity}$ = .001; women: RR = 1.30, 95% CI 1.11-1.53; $P_{heterogeneity}$ = .001).

Sensitivity Analysis

Because of the high heterogeneity, we conducted a sensitivity analysis. Eliminating one study in each group did not predominantly change the overall RR (Figs S1-S3), which indicates the reliability of our results, despite some high heterogeneity.

Publication Bias

Publication bias was detected by Begg's and Egger's linear test and summarized in Table 3. Neither test for the HZO infection and stroke association indicated publication bias, but both tests indicated publication bias for the HZ infection and MI association. However, the pooled point estimation and 95% CI did not change after the trim-and-fill method (Table S2), so the meta-analysis results for HZ infection and MI remained stable despite publication bias.

Discussion

This meta-analysis of 11 observational studies demonstrated that risk of stroke/TIA was increased with HZ

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Table 1. *Summary of basic characteristics*

(Ref.) Author (year)	Country	Design	Study period	Follow-up (years)	No of participants	Age	Infection	Endpoints	Positive/Negative	NOS score
(8) Minassian et al (2015)	USA	Case–control	2006-2011	5 (median)	42,954	80 (median)	HZ	Stroke	Positive	8
(8) Minassian et al (2015)	USA	Case–control	2006-2011	5 (median)	24,237	80 (median)	HZ	MI	Negative	8
(8) Minassian et al (2015)	USA	Case–control	2006-2011	5 (median)	6,971	80 (median)	HZO	Stroke	Positive	8
(16) Calabrese et al (2017)	USA	Cohort	2006-2013	8	43,527	71 (mean)	HZ	Stroke	Positive	7
(17) Kwon et al (2016)	Korea	Cohort	2003-2013	11	766,179	39 (median)	HZ	Stroke	Positive	7
(18) Sundstrom et al (2015)	Sweden	Cohort	2008-2010	1	13,296	0+	HZ	Stroke	Positive	7
(19) Sreenivasan et al (2013)	Denmark	Cohort	1995-2008	13	4,620,980	18+	HZ	Stroke	Positive	7
(20) Lin et al (2010)	Taiwan	Cohort	2003-2004	1	658	56.9 (mean)	HZO	Stroke	Positive	9
(21) Kang et al (2009)	Taiwan	Cohort	1997-2001	1	7,760	46.7 (mean)	HZ	Stroke	Positive	9
(21) Kang et al (2009)	Taiwan	Cohort	1997-2001	1	120	46.7 (mean)	HZO	Stroke	Positive	9
(22) Breuer et al (2014)	UK	Cohort	2002-2010	6.3 (median)	106,601	59.4 (median)	HZ	Stroke/TIA	Negative	9
(22) Breuer et al (2014)	UK	Cohort	2002-2010	6.3 (median)	106,601	59.4 (median)	HZ	MI	Positive	9
(22) Breuer et al (2014)	UK	Cohort	2002-2010	6.3 (median)	1,710	59.4 (median)	HZO	Stroke	Negative	9
(23) Elkind et al (2016)	International	Case–control	2010-2014	5	326	7.7 (mean)	HZ	Stroke	Negative	7
(24) Yawn et al (2016)	USA	Cohort	1986-2011	7 (mean)	4,478	68.2 (mean)	HZ	Stroke	Negative	9
(24) Yawn et al (2016)	USA	Cohort	1986-2011	7 (mean)	4,454	68.2 (mean)	HZ	MI	Positive	9
(25) Langan et al (2014)	UK	Case–control	1987-2012	12.5 (median)	6,584	77 (median)	HZ	Stroke	Negative	7
(25) Langan et al (2014)	UK	Case–control	1987-2012	12.5 (median)	6,584	77 (median)	HZO	Stroke	Negative	7

Abbreviations: HZ, herpes zoster; HZO, herpes zoster ophthalmicus; MI, myocardial infarction; NOS: Newcastle-Ottawa quality assessment scale; TIA, transient ischemic attack.

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Table 2. *Summary of meta-analysis results*

Infection	Endpoint	Subgroup	No. of studies	Model	RR (95% CI)	I^2 (%)	$P_{heterogeneity}$
HZ	Stroke		13	Random	**1.30 (1.17, 1.46)**	81.9	.001
	Stroke type	Ischemic	5	Fixed	**1.22 (1.07, 1.40)**	49	.097
		Hemorrhagic	3	Random	1.52 (.88, 2.63)	88.4	.001
	Time since HZ	<1 month	4	Random	**1.92 (1.47, 2.51)**	95.3	.001
		1-3 months	2	Random	**1.27 (1.05, 1.53)**	79.8	.026
		3-12 months	6	Fixed	1.02 (.99, 1.04)	45.5	.102
		>12 months	2	Fixed	**1.05 (1.02, 1.09)**	0	.623
	Area	Europe	10	Random	**1.23 (1.13, 1.34)**	91.0	.001
		America	8	Random	**1.28 (1.07, 1.53)**	98.4	.001
		Asia	2	Random	**1.60 (1.11, 2.30)**	91.9	.001
	Age	<40	8	Random	**2.03 (1.64, 2.51)**	77.8	.001
		40-59	6	Random	**1.35 (1.20, 1.53)**	89.1	.001
		>60	9	Random	**1.32 (1.17, 1.48)**	95.9	.001
	Sex	Male	5	Random	**1.31 (1.09, 1.57)**	87.4	.001
		Female	5	Random	**1.30 (1.11, 1.53)**	86.9	.001
	Treatment	Yes	6	Fixed	**1.17 (1.08, 1.27)**	0	.768
		No	6	Random	**1.38 (1.06, 1.80)**	84.5	.001
HZ	MI		6	Random	**1.18 (1.07, 1.30)**	90.2	.001
HZO	Stroke		8	Random	**1.91 (1.32, 2.76)**	84.4	.001

Abbreviations: CI, confidence interval; HZ, herpes zoster; HZO, herpes zoster ophthalmicus; RR, relative risk.
The bold numbers mean statistical significance with its confidence interval without 1.

Table 3. *The results of publication bias*

	Egger's *P* value	Begg's *P* value
HZ and stroke	.263	.536
HZ and MI	.014	.035
HZO and stroke	.059	.373

Abbreviations: HZ, herpes zoster; HZO, herpes zoster ophthalmicus; MI, myocardial infarction.

infection, even when controlling for multiple risk and confounding factors. The association of HZ infection and MI was less strong but still significant. We observed a strong effect for risk of stroke after HZO diagnosis, with a 1.9-fold increased risk.

Several possible biological stroke mechanisms are invoked with the association of risk of stroke/TIA or MI after HZ infection. First, inflammation associated with systemic infection may injure the endothelium and lead to arterial thrombosis with the background of atherosclerosis and hypercoagulability.[34,35] Second, reactivated VZV travels along nerve fibers in the trigeminal ganglion and damages the vessel wall, which leads to aneurysm and hemorrhage.[36-38] In addition, patients with HZ often have postherpetic neuropathic pain, which induces depressive symptoms and a weakened immune system, to increase the risk of stroke.[39,40]

On subgroup analysis, HZ infection was associated with increased risk of ischemic stroke but not hemorrhagic stroke. An explanation might be the complete difference in pathological changes and etiology between ischemic stroke and hemorrhagic stroke. The etiology of ischemic stroke is mostly caused by inflammation and HZ infections are correlated with other major comorbidities, such as diabetes, cardiac disease, and hypertension,[41-43] which is favorable for ischemic than hemorrhagic stroke. Insufficient studies for statistical power might explain the RR of hemorrhagic stroke with no statistical significance. Further exploration into this matter was suggested. We found the greatest increase in risk for stroke at 1 month after HZ diagnosis, with a 1.9-fold increased risk, then 1-3 months and gradually decreasing with time, which indicates the need for early preventive measures. As well, patients less than 40 years old had a twofold increased risk of stroke after HZ diagnosis, with a threefold increased risk for those 40-59 and higher than 60 years old. This finding may be explained in part by the voluntary screening of vascular risk factors in patients older than 45 years in some countries such as the United Kingdom.[44,45] In addition, VZV is a risk factor for stroke in people less than 40 years old, which was supported by several studies.[22,46] With the large samples in our analysis, we found no difference between men and women in stroke risk after HZ but did find that the risk differed by region, with different geographical and climatic situations, lifestyles, working habit, and genotypic milieu. Asian populations had a 1.6-fold increased risk of stroke after HZ exposure, which suggests that this group should receive more attention.[47,48]

Whether antiviral therapy alters the risk of stroke after HZ exposure is an issue. In our subgroup analysis, the risk of stroke was lower with treatment than without treatment,

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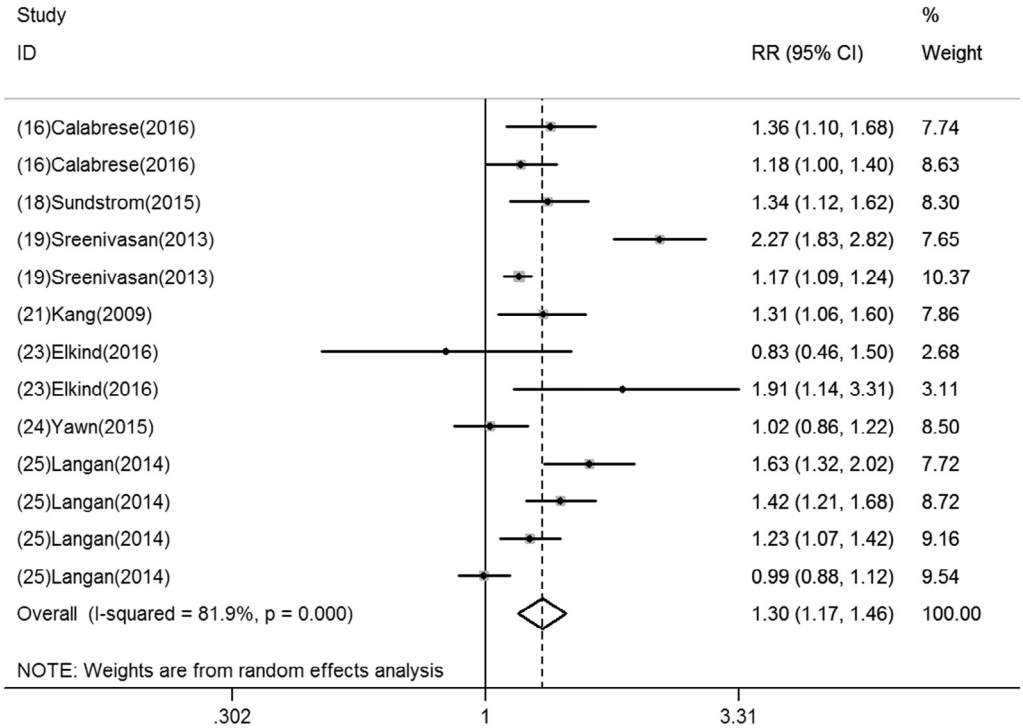

Figure 2. *Forest plot of stroke/transient ischemic attack risk with herpes zoster. The effect was considered statistically significant with 95% CI for the overall RR not overlapping 1. When the study provided results by subgroup of time since herpes zoster but no overall RR estimate, the subgroup RRs were used. The referred studies of 16, 19, 23, and 25 did not provide the overall RR, the subgroup RRs were used. Abbreviations: CI, confidence interval; RR, relative risk.*

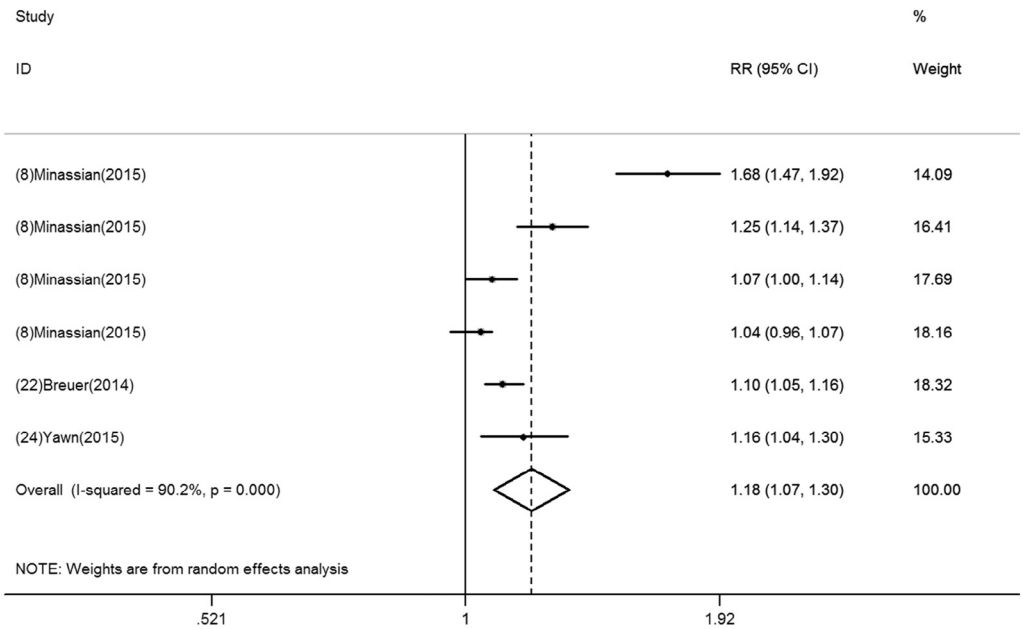

Figure 3. *Forest plot of myocardial infarction risk with herpes zoster. The effect was considered statistically significant with 95% CI for the overall RR not overlapping 1. When the study provided results by subgroup of time since herpes zoster but no overall RR estimate, the subgroup RRs were used. The referred studies of 8 did not provide the overall RR, the subgroup RRs were used. Abbreviations: CI, confidence interval; RR, relative risk.*

although not significantly. Previous studies demonstrated that the actual difference in risk should be greater than what we obtained. We found that people who received antiviral therapy more frequently had severe disease, which would indicate confounding and reduce the advantage of antiviral therapy.[8,25]

Our study has several strengths. First, we evaluated the risk of stroke/TIA with HZ infection but also the risk of

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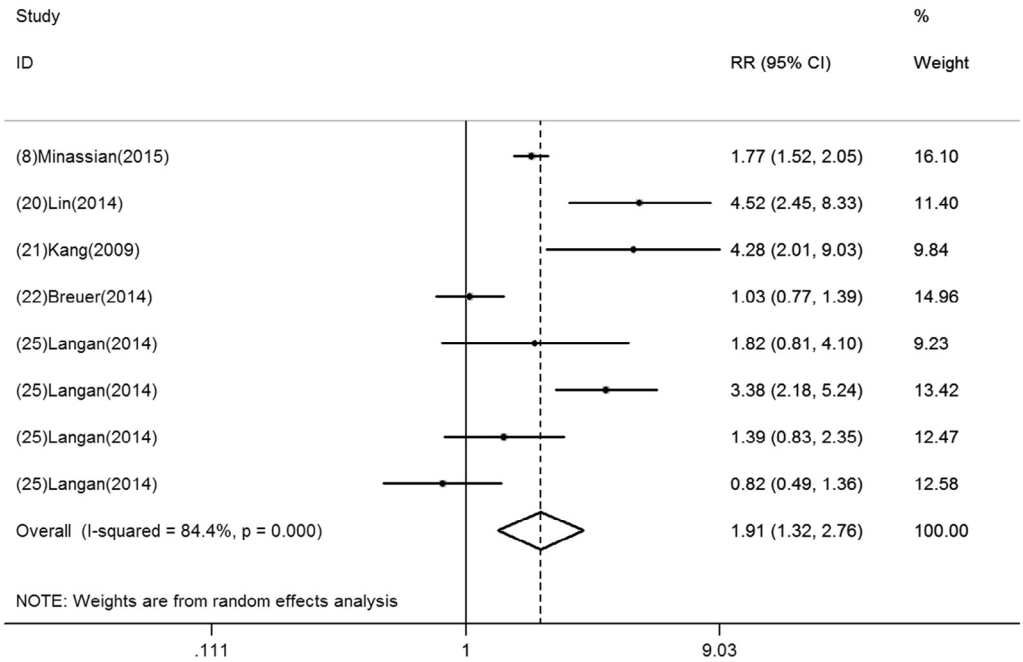

Figure 4. *Forest plot of stroke risk with herpes zoster ophthalmicus. The effect was considered statistically significant with 95% CI for the overall RR not overlapping 1. When the study provided results by subgroup of time since herpes zoster but no overall RR estimate, the subgroup RRs were used. The referred studies of 25 did not provide the overall RR, the subgroup RRs were used. Abbreviations: CI, confidence interval; RR, relative risk.*

MI as well as the risk of stroke with HZO infection. Our sensitivity analysis revealed that all 3 overall analyses were robust. Publication bias analyses also revealed that our results were stable. Second, all of our included articles were high quality and included large samples, which guaranteed the degree of accuracy and validity in our meta-analysis. For most cohort studies in our review, long-term follow-up with a high follow-up rate was a prominent advantage. Third, we performed several subgroup analyses to lower the heterogeneity and have more robust and accessible results, by investigating time since infection, sex, age, region, and treatment. We found the risk of stroke with HZ infection lower with treatment than without treatment, which strengthens the association between HZ infection and stroke risk and suggests that patients should receive antiviral therapy after HZ infection as soon as possible to prevent stroke.

Our study has several limitations. First, most of our included studies used administrative data to identify HZ, which could suggest 5%-15% false-positive cases because the ICD code may not be accurate.[49] However, we considered TIA as minor stroke and the code for TIA diagnosis was likely to be incorrect.[1,50] Meanwhile, some silent strokes or MI may not have been recorded.[24] Second, we inevitably observed the high heterogeneity in our studies because of the study design, methodological differences, geographic regions, confounding factors, and the inherent differences between individuals, although we used the random effects model. Third, we found publication bias in the HZO analysis, which may be related to the search strategy restricted to English language literature.

In conclusion, our systematic review and meta-analysis revealed increased risk of stroke/TIA or MI with HZ infection and HZO as an important risk factor for stroke. HZ infection was associated with increased risk of ischemic stroke but not hemorrhagic stroke. The risk of stroke after HZ infection was reduced with time since infection. Further studies are needed to explore whether zoster vaccination could reduce the risk of stoke/TIA or MI and how to prevent HZ infection initially.

Appendix: Supplementary material

Supplementary data to this article can be found online at doi:10.1016/j.jstrokecerebrovasdis.2017.04.013.

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